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Roll No

FT-8001 (CBGS)

B.E. VIII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Industrial Hygiene and Occupational Health

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is industrial ventilation? Explain in brief the purpose of a ventilation system. 7
b) A recently graduated industrial Hygienist is assigned to calculate TWA for an employee who is exposed to 160 ppm of Toluene for 25 minutes out of each hour and during remaining period of each hour there is exposure of 55 ppm of Toluene during work shift of 8 hours? What should he report? 7
2. a) Explain in brief Ionizing and Non-ionizing radiations and their adverse health effects. 7
b) Calculate daily noise dose for an employee who has been exposed for followings levels of sound during a work shift of 8 hours. 7
 - i) 95 dB for 30 minutes
 - ii) 115 dB for 90 minutes
 - iii) 90 dB for 2 hours
 - iv) 85 dB for 2 hours
 - v) 105 dB for 1 hour
 - vi) 90 dB for 1 hour

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3. a) Explain Ergonomics and its applications for safety and Health. 7
b) Explain in detail "work physiology" of a worker in work place. 7
4. a) Discuss in detail occupational disease "Silicosis" and its preventive measures. 7
b) Explain in brief the fundamentals of first-aid. 7
5. a) Write the specification for head protection equipment. 7
b) Describe in brief respiratory personal protective devices. 7
6. a) An employee is exposed to ammonia concentration of 65ppm for first 4 hours in the shift of 8 hours and for remaining every hour he is exposed to 30 ppm. Calculate TWA and access whether it is exceeded. 7
b) Calculate combined sound pressure level in dB of six sound sources having individual pressure level of 91, 96, 103, 115, 114 and 92 dB at particular location. 7
7. a) How the applications of ergonomics help in safety and health of an employee? 7
b) Brief in detail about the compensation of occupational diseases in an industry. 7
8. a) Explain the in brief classification of respiratory hazards. 7
b) What do you mean by musculoskeletal disorders (MSDs)? How are work postures and movements a risk for MSDs? 7

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